

AI in Culture and Arts – Human-AI Interaction

Introduction to Artificial Intelligence and Interactive Machine Learning

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München

1. How to Model Intelligence?

2. When is Learning Possible?

3. How Do Machines Learn?

4. How Do Humans Train Machines?

5. How Do Humans Teach Machines?

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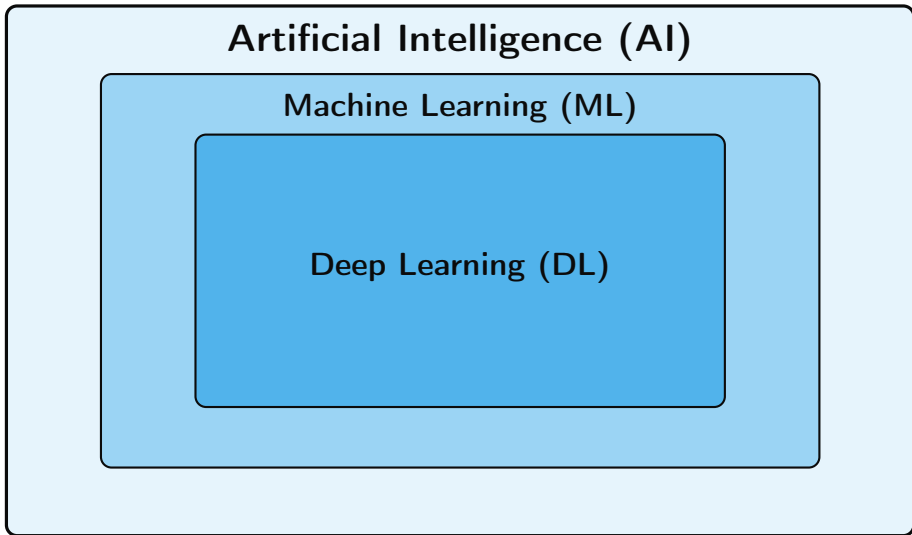
5. How Do Humans Teach Machines?

“The ability of an agent to achieve goals in a wide range of environments.”
– (Russell, 2019)

Let E be the space of all computable reward summable environmental measures with respect to the reference machine $\mathcal{U}$, and let K be the Kolmogorov complexity function. The expected performance of agent π with respect to the universal distribution $2^{-K(\mu)}$ over the space of all environments E is given by,

$$\Psi(\pi) := \sum_{\mu \in E} 2^{-K(\mu)} V_{\mu}^{\pi}.$$

We call this the **universal intelligence** of agent π (Legg & Hutter, 2007).



Most of the hype is based on **Deep Learning**.

Van Rooij et al. describe how the usage of the term “AI” shifted from *AI-as-Psychology* to *AI-as-Engineering* (van Rooij et al., 2024).

Historical (1970s–80s)	Contemporary (Today)
<i>AI-as-Psychology</i> Goal: Understand the mind. AI as a theoretical tool .	<i>AI-as-Engineering</i> Goal: Technical progress. AI as a product .

My critique of the *AI-as-Engineering* view is the following:

- Environment and agent (the system) are co-defined and evolve together in an evolutionary drift. There is no agent without an environment and vice versa.
- There is no necessary reason why a process that generates something surprising and/or coherent to us requires thinking.
- There is no necessary reason that a *theory of cognition as computation* implies that it is practically feasible to recreate a human-level mind in a machine (van Rooij et al., 2024).
- Any performance metric is made up by “linguaging” beings in a certain social context.
- The “goal” organisms seem to follow is just to continue their *autopoiesis* (Maturana & Varela, 1980) and the only reason why they *exist* is because they do so. Therefore, there is no goal but only the condition of the organism's existence.

Therefore, I think it would be helpful to

- think of large language models (LLMs) not as intelligent but as being able to **participate in communication** (Esposito, 2022).
- see them as **social-cultural technology** (Farrell, Gopnik, Shalizi, & Evans, 2025) and new medium that **couples psychic and social systems** (Zönnchen, Dzhimova, & Socher, 2025).
- to focus more on **a shift in communication** similar to the invention of the printing press which will co-evolve with minds.

“Cybernetics realizes the Hegelian logic of reflection: in cybernetics, reflection—the circularity of consciousness reflecting on itself—becomes a real, material process transferred into machines.” – (Hui, 2024)

Symbolism

Exploits explicit, rule-based symbolic manipulation, logic, and structured reasoning to represent knowledge and solve problems.

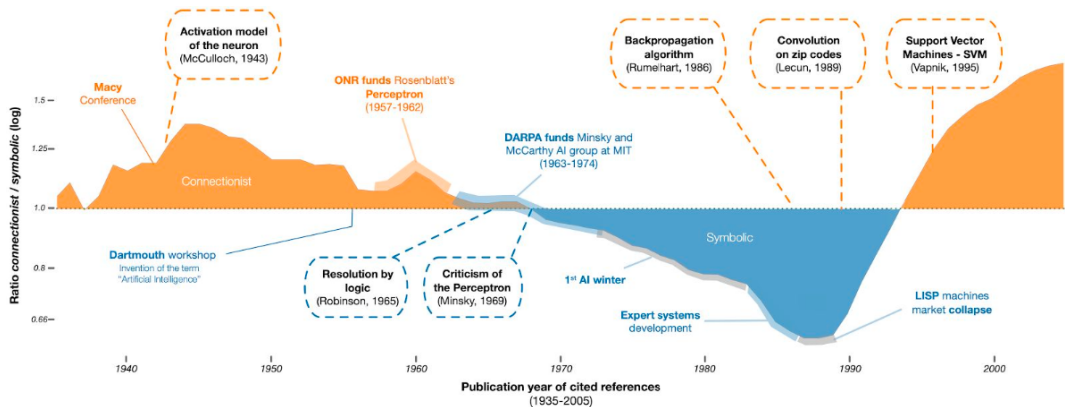
- **Assumption:** Intelligence uses high-level, human-readable symbols to represent problems and logic to solve them.
- **Motivation:** Model the **mind** (as a rational unit)!

Connectionism

Exploits artificial neural networks & statistics, emphasizing learning from patterns, distributed representations, and emergent behaviors.

- **Assumption:** Intelligence emerges from the interaction of simple and low-level units, i. e. biological neurons.
- **Motivation:** Model the **brain** (and the mind will emerge)!

How to Model Intelligence?



Source: (Cardon et al., 2018)

Let's say we know that "if it rains R , the grass is wet W ."

$$R \rightarrow W$$

Then if it rains ($R = \text{true}$), we can deduce that the grass is wet ($W = \text{true}$)!

We call this a *rule of inference* and this particular example is called *modus ponens*.

In logic, *rules of inference* are ways of deriving conclusions from premises (deduction). If an argument with true premises follows a rule of inference then the conclusion cannot be false.

Exercise (Deduction)

Suppose you have (finitely) many

1. rules $A \rightarrow B, B \rightarrow C, D \rightarrow E, A \rightarrow E, \dots$, and
2. (true) premisses (“facts”) $A, D, H, \dots$

How can you deduce all “facts”?

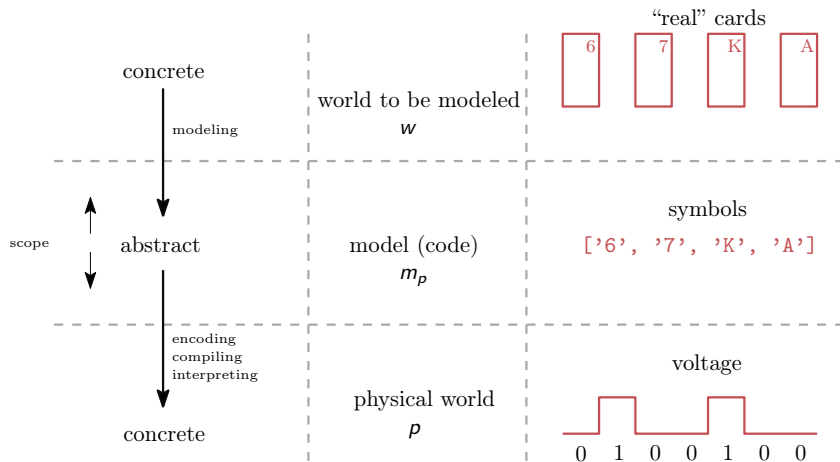
We could deduce that “raining” ($R = \text{true}$) implies the “wetness of the grass” ($W = \text{true}$) even if we have no idea of what rain or wetness is—how it feels like, how it happens, and what it does, etc.

We only need the correct rules and premisses. What is left is symbol manipulation!

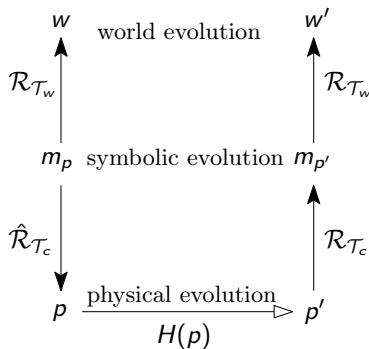
⇒ a computer can do it!

⇒ **Assumption:** To get a model of intelligence we “just” need to put in all the rules and premisses into a machine.

What is physical computation?



What is physical computation?



1. theory $\mathcal{T}$
2. representation relation $\mathcal{R}_{\mathcal{T}}$ in relation to a theory $\mathcal{T}$
3. reverse representation relation $\hat{\mathcal{R}}_{\mathcal{T}}$

Assumptions of AI Research of the Time

- **The biological assumption:** The brain processes information in discrete operations by way of some biological equivalent of on/off switches.
- **The psychological assumption:** The mind can be viewed as a device operating on bits of information according to formal rules.
- **The epistemological assumption:** All knowledge can be formalized.
- **The ontological assumption:** The world consists of independent facts that can be represented by independent symbols.

Dreyfus attacks all of these assumptions in (Dreyfus & Dreyfus, 1986; Dreyfus, 1965) using the (European) tradition of phenomenology (Husserl, Heidegger, Merleau-Ponty).

“What does [Dreyfus] offer us? Phenomenology! That ball of fluff. That cotton candy!” – Edward Feigenbaum

"[Dreyfus'] derisiveness has been so provoking that he has estranged anyone he might have enlightened. And that's a pity." – Pamela McCorduck

Video

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- **2006–2012: Early Breakthroughs**
 - Revival of deep neural networks combining backpropagation and gradient descent (Leibniz, 1676; Rosenblatt, 1962; Pontryagin, 1950; Dreyfus, 1962; Amari, 1967; Linnainmaa, 1970; Werbos, 1982; Rumelhart, 1982; LeCun, 1987).
 - Major success: ImageNet competition (AlexNet, 2012)
- **2013–2017: Rapid Expansion of Deep Learning**
 - CNN dominance in image recognition
 - Reinforcement Learning breakthroughs (AlphaGo, 2016)
- **2018–Present: Transformers and Large Language Models**
 - Natural Language Processing revolution (BERT, GPT)
 - Generative AI models (GPT-3, ChatGPT, Stable Diffusion)
- **Modern Symbolic Revival and Hybrid AI (Ongoing)**
 - Neurosymbolic AI: Combining deep learning and symbolic reasoning
 - Explainable AI (XAI): Interpretable, trustworthy systems
 - Importance for ethics, transparency, and safety

1. Computational Power

- Massive increase in GPU computing capacity.
- Parallelization of neural network training.

2. Big Data

- Availability of huge, labeled datasets.
- Online data collection (ImageNet, web scraping).

3. Algorithmic Advances

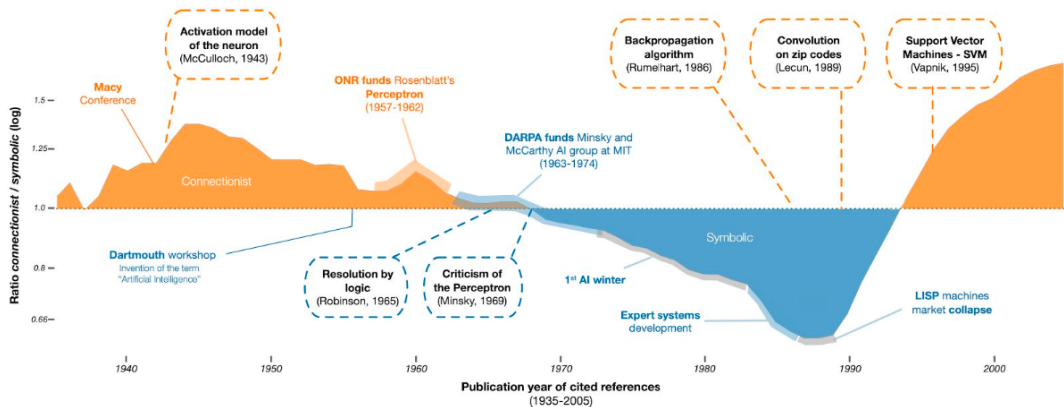
- Improved training algorithms (e. g., backpropagation, Adam optimizer).
- New network architectures (CNNs, RNNs, Transformers).

4. Open-source Ecosystem

- Accessible tools: TensorFlow, PyTorch.
- Community-driven knowledge sharing.

Video

How to Model Intelligence?



Source: (Cardon et al., 2018)

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“The organism feeds on negative entropy.” – (Schrödinger, 1944)

0101010101010101010101010101

110011010101110100011100101111

Let's say we have the following two 30-bit information:

$$x_1 = 0101010101010101010101010101$$

$$x_2 = 110011010101110100011100101111$$

Question: Which carries “more” information?

Answer: We say the **entropy** (Shannon, 1948) of x_2 is higher than the entropy of x_1 . If we read the bits from left to right we are more “often” **surprised** when reading x_2 .

Information entropy measures uncertainty / surprise. It is the **expected value of surprise**.

- High entropy: highly unpredictable, many possible outcomes, each similarly likely
- Low entropy: predictable, only a few likely outcomes

Learning requires

$$\text{entropy} + \text{structure} \approx \text{complexity}.$$

We have to be surprised but also be able to exploit a structure to compress observations.

A physical formula like

$$F = m \cdot a$$

can be thought of as a **highly compressed representation** of some aspects of physical reality!

In other words, $F = m \cdot a$ reveals the structure of motion, thus the informational entropy cannot be too high.

Some neuroscientists (K. Friston, Kilner, & Harrison, 2006; K. J. Friston, 2011) think that organisms try to **minimize surprise** by

1. adjusting expectations (perception, learning) and
2. realize expectations (acting).

According to this school of neuroscience,

organisms are their own existence proof.

By acting to keep themselves alive, they continuously generate sensory inputs that confirm their continued existence.

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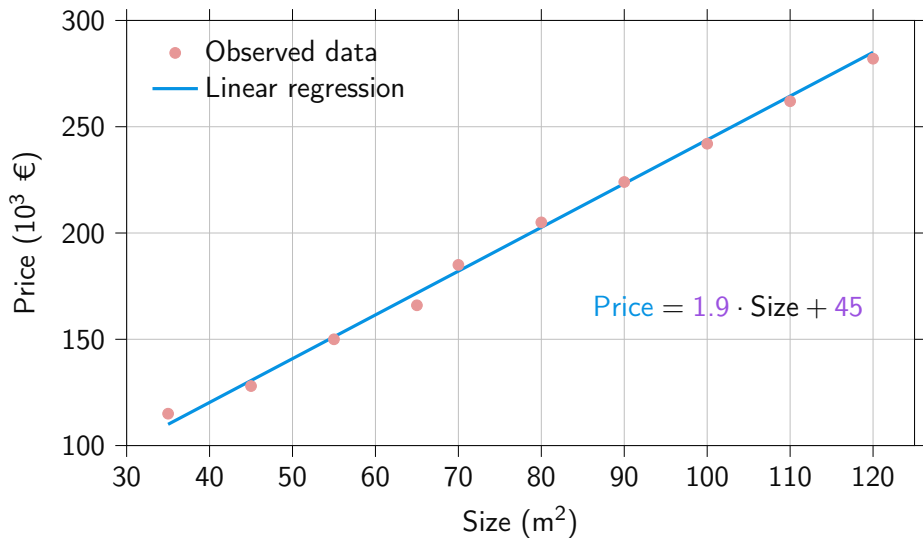
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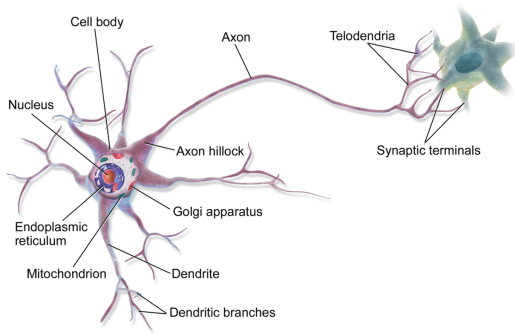
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*"We do not learn from experience [...] we learn from reflecting on experience."
– John Dewey*

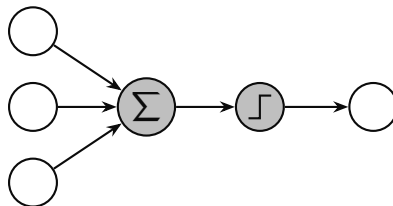
Learning a model of “the world”



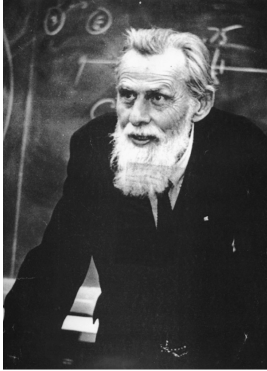
Neuron



Artificial neuron

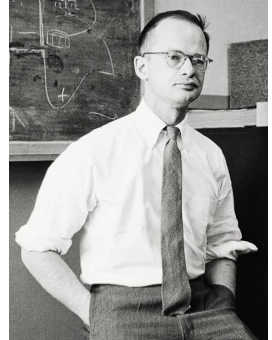
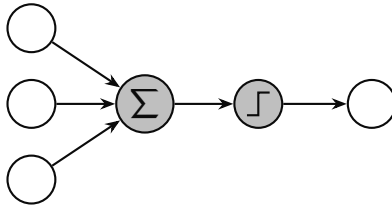


The Mathematical Neuron



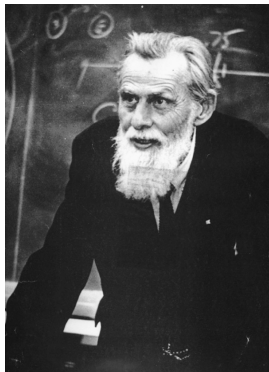
Warren S. McCulloch

"Because of the 'all-or-none' character of nervous activity, neural events and the relations among them can be treated by means of propositional logic."



Walter H. Pitts Jr.

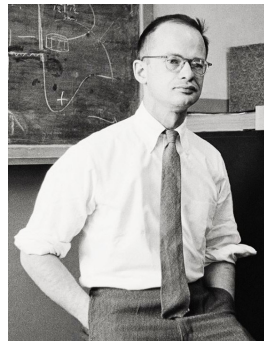
The Mathematical Neuron



Warren S. McCulloch

"Because of the 'all-or-none' character of nervous activity, neural events and the relations among them can be treated by means of propositional logic."

$$f(x_1, \dots, x_n) = \begin{cases} 1 & \text{if } \sum_{k=1}^n x_k > 1 \\ 0 & \text{otherwise.} \end{cases}$$

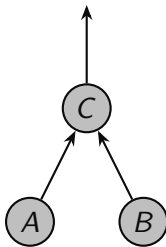


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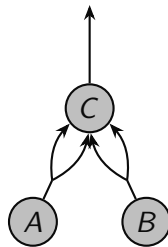
Complex logical operations can be performed using networks of binary neurons.



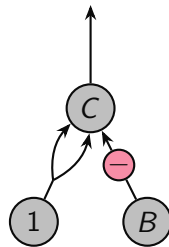
Identity: $C = A$



And: $C = A \wedge B$

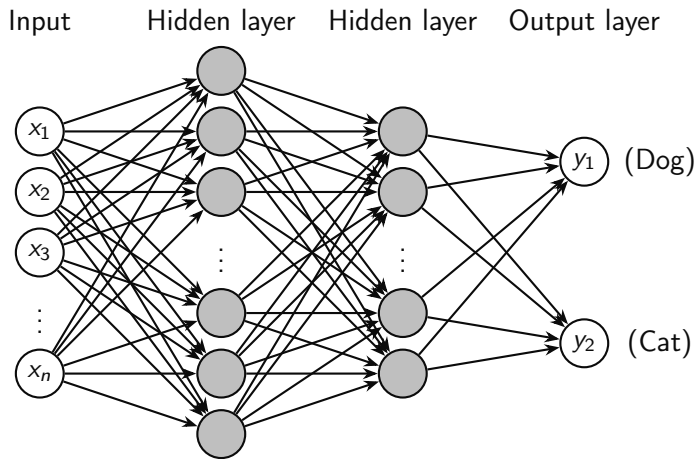


Or: $C = A \vee B$



Negation: $C = \neg B$

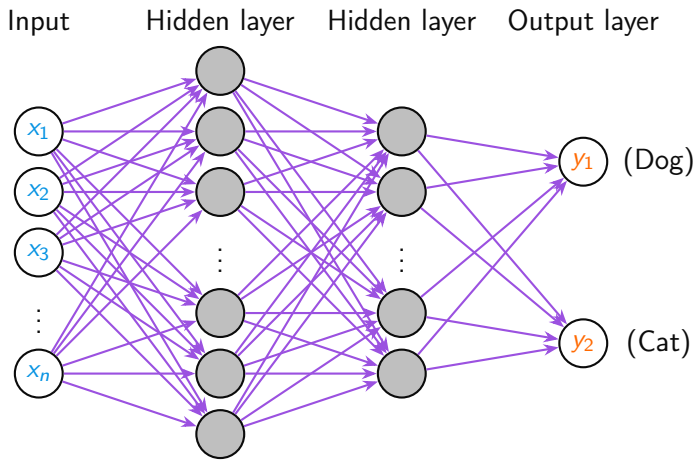
Synaptic Plasticity



$$h_{\theta}(\mathbf{x}) = \mathbf{y},$$

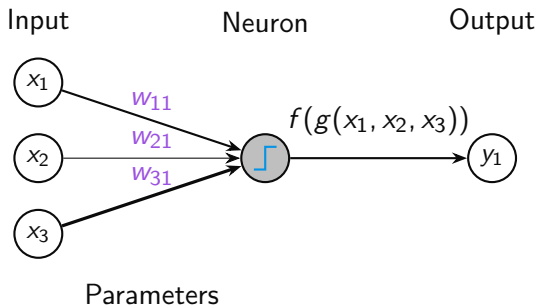
where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_k)$

Synaptic Plasticity



$$h_{\theta}(\mathbf{x}) = \mathbf{y},$$

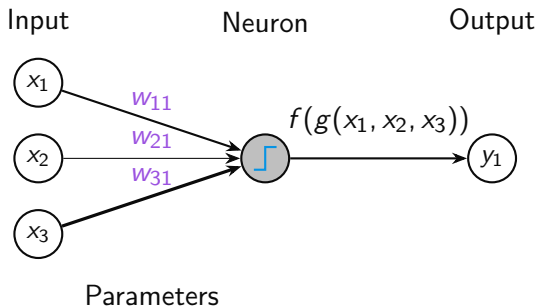
where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_k)$



Parameters determine how strong neurons are wired together:

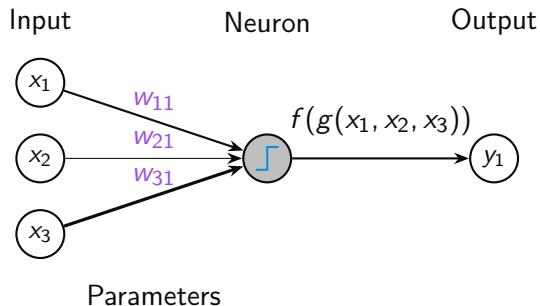
$$g(x_1, x_2, x_3) = x_1 \cdot w_{11} + x_2 \cdot w_{21} + x_3 \cdot w_{31}$$

Synaptic Plasticity



"Neurons that fire together, wire together."

$$w_{ij} = w_{ij} - \eta \cdot x_i \cdot y_j$$

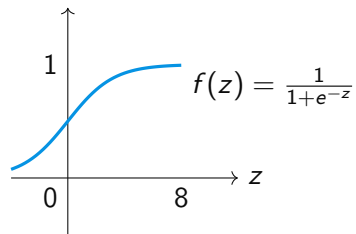
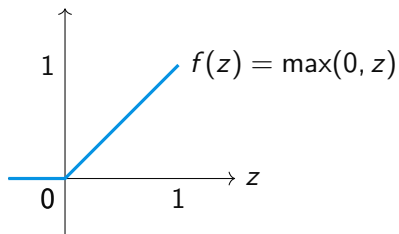
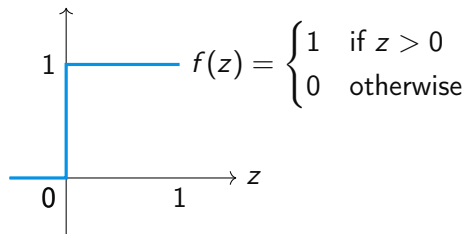
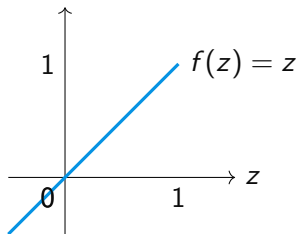


"Neurons that fire together, wire together."

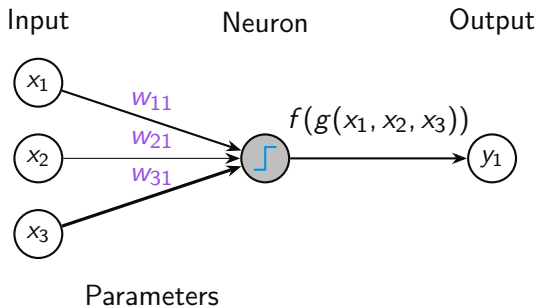
$$\theta_{t+1} = \theta_t - \eta \cdot \nabla J(\theta_t)$$

In this case $\theta_t = (w_{11}, w_{21}, w_{31})$.

Activation Functions



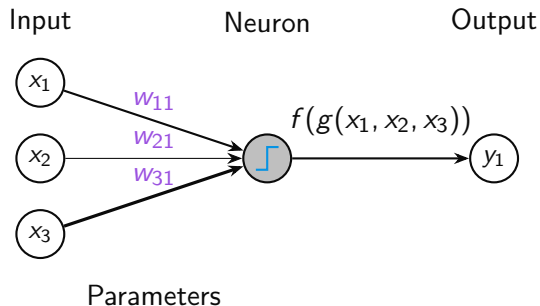
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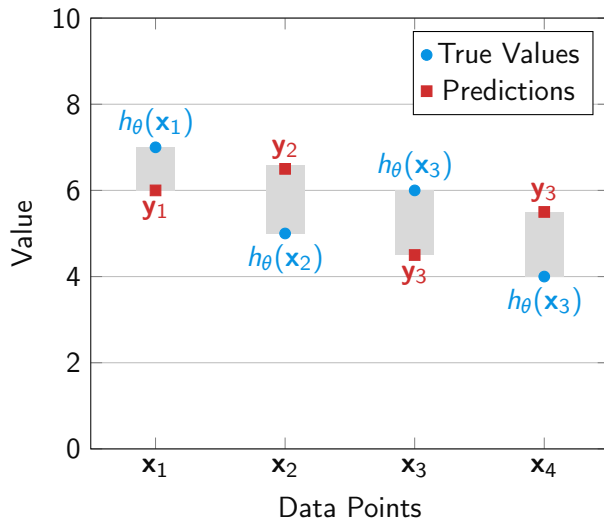
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Cost Function (Regression)

Mean Squared Error (MSE):



Mean Squared Error (MSE):

$$J(\theta) = \frac{1}{N} \sum_{i=1}^N (\mathbf{y}_i - h_{\theta}(\mathbf{x}_i))^2$$

where $\mathbf{y}_i$ is the correct label of a data point $\mathbf{x}_i = (x_1, \dots, x_n)$ in our training data.

Example: Let us suppose

$$h_{\theta}(x_1, x_2) = w_{11} \cdot x_1 + w_{21} \cdot x_2$$

and let us assume $w_{11} = 2$, $w_{21} = 0.5$ and we have two data points $\mathbf{x}_1 = (1, 1)$, $y_1 = 1$ and $\mathbf{x}_2 = (-1, -2)$, $y_2 = -3$. Then our mean squared error is:

$$\begin{aligned} J(w_{11}, w_{21}) &= \frac{1}{2} [(1 - (w_{11} \cdot 1 + w_{21} \cdot 1))^2 + (-3 - (w_{11} \cdot (-1) + w_{21} \cdot (-2)))^2] \\ &= \frac{1}{2} [(1 - (2 \cdot 1 + 0.5 \cdot 1))^2 + (-3 - (2 \cdot (-1) + 0.5 \cdot (-2)))^2] \end{aligned}$$

The gradient would be:

$$\nabla J(w_{11}, w_{21}) = \begin{bmatrix} -4 + 2w_{11} + 3w_{21} \\ -7 + 3w_{11} + 5w_{21} \end{bmatrix}$$

Categorical Cross Entropy Cost:

$$J(\theta) = -\frac{1}{N} \sum_{i=1}^N [\mathbf{y}_i \cdot \log(h_{\theta}(\mathbf{x}_i))]$$

where $\mathbf{y}_i$ is interpreted as the probability distribution of categories for $\mathbf{x}_i = (x_1, \dots, x_n)$, i.e. a data point.

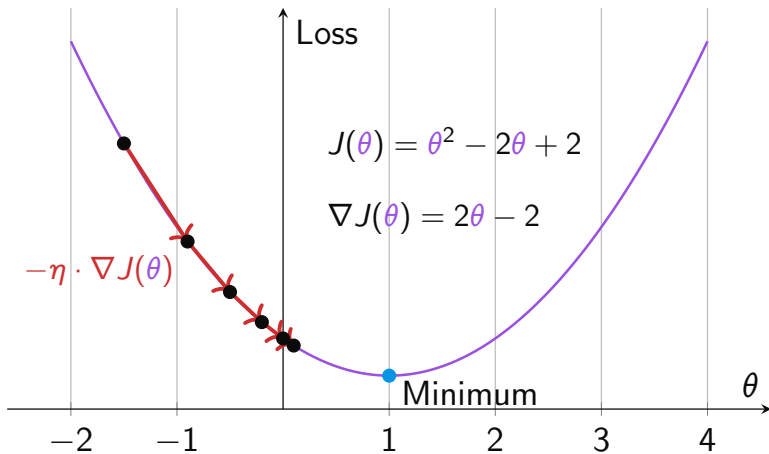
To improve the model's prediction, we try to minimize the cost function. One way to do this is **gradient descent**:

$$\theta_{t+1} = \theta_t - \eta \cdot \nabla J(\theta_t)$$

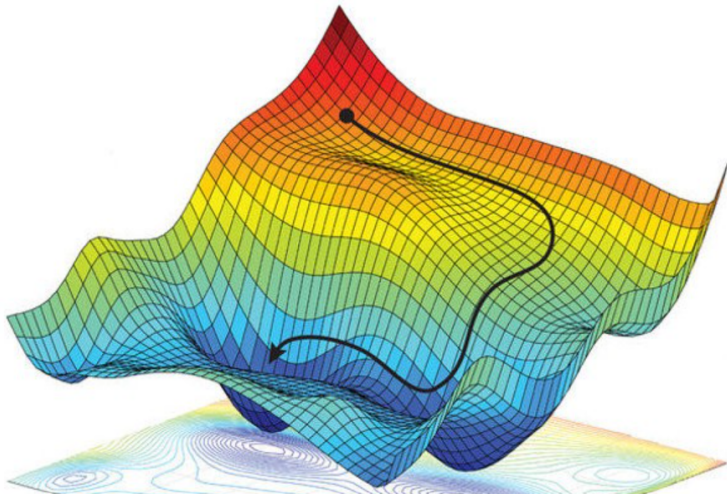
Condition: $\nabla J(\theta_t)$ exists!

Interactive Tutorial

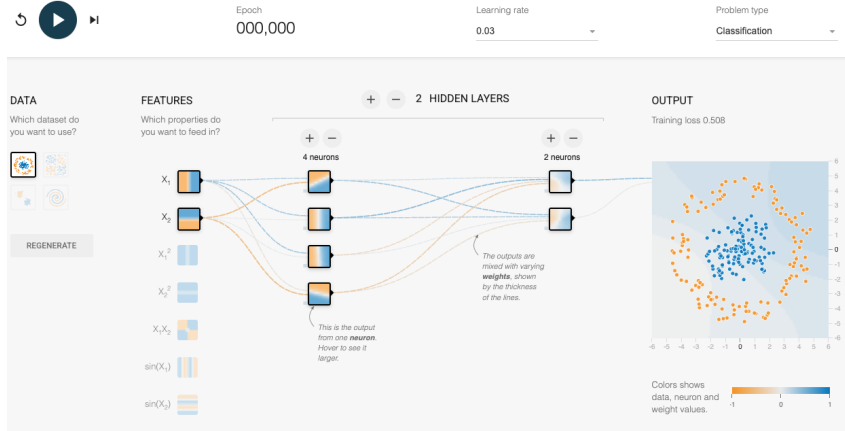
Gradient Descent



Gradient Descent



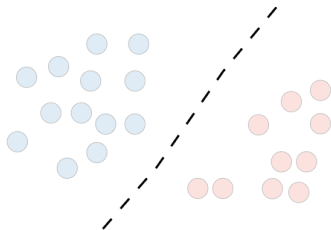
Design and Try Your Perceptron



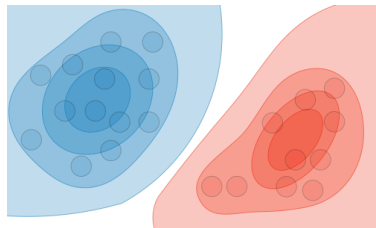
Simplified Tensorflow Playground

Extended Tensorflow Playground

- **Discriminative models:** Learn the boundaries of decisions.
- **Generative models:** Learn the whole distribution of the data.



Discriminative modelling



Generative modelling

Artificial neural networks (ANNs) are the most prominent model types currently in use. However machine learning spans over much more model types such as:

Model type	Best for	Complexity
Linear models	Simple trends, high interpretability	Low
Tree-based	Tabular data, excel-style sheets	Medium
SVM	Clear margins of separation	Medium
Bayesian	Risk assessment, small datasets	Low/Medium
Neural networks	Images, audio, large-scale text	High

Table: Comparison of machine learning model types.

The “AI-Hype” Models

- **Scale:** Trillions of parameters (LLMs, GenAI).
- **Compute:** Requires data centers and GPU clusters.
- **Latency:** High ($> 1s$ common); depends on API/Internet.
- **Agency: Low.** The user is a consumer of the output.
- **Goal:** General intelligence, content generation.

Interactive & Local ML

- **Scale:** Small models and training data set.
- **Compute:** Runs locally on a standard laptop CPU. Often trainable on a single GPU/CPU.
- **Latency: Real-time** ($< 10ms$). Essential for user loops.
- **Agency: High.** The user steers the model via rapid feedback loops.
- **Goal:** Gesture control, active learning, privacy, real-time interaction.

Geoffrey Hinton on AI and consciousness:

Video

Yann LeCun on the need for a paradigm shift:

Video 1

Video 2

1. How to Model Intelligence?

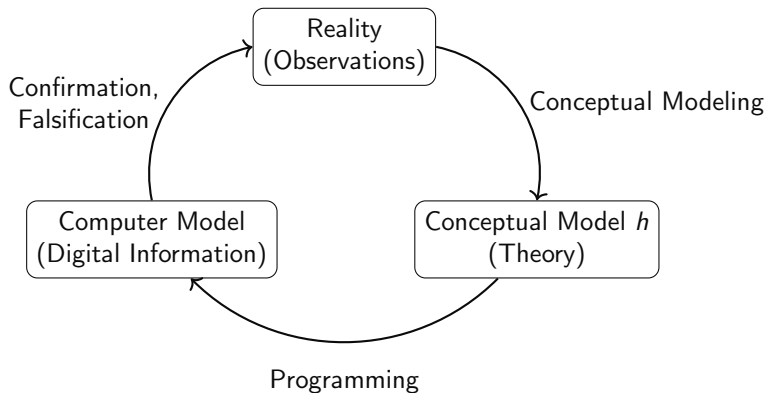
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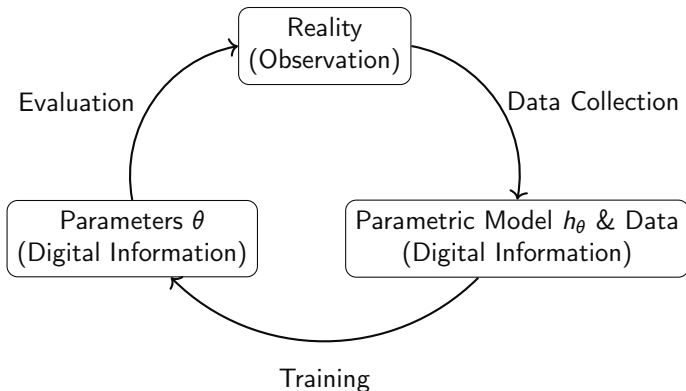
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Theory-driven Modeling

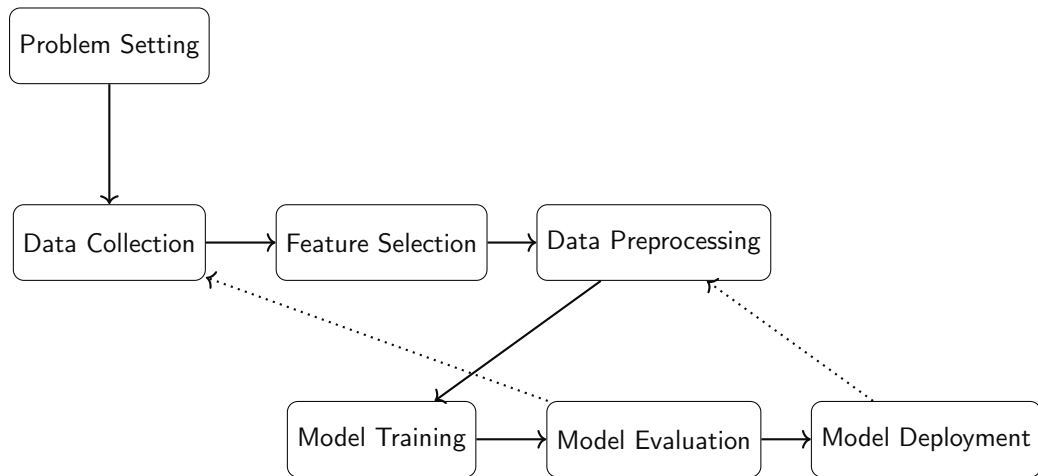


Minds constructs a (falsifiable) theory or hypothesis about reality to test against.



Algorithms (directly) fit a parametric model to the data. **Minds** are usually unable to conceptualize the trained model.

Development Cycle



```
100 class fast_qlinear(torch.autograd.Function):
101     def forward(ctx, a, b, scales, zeros):
102
103         m, k = a.shape
104         _, n = b.shape
105
106         quant_groupsize = 128
107         block_size_m = 16
108         block_size_n = 32 # [N = 4096 // 32] = 128 blocks
109         block_size_k = 256
110         group_size_m = 8
111         num_warps = 4
112         num_stages = 8
113         total_blocks_m = triton.cdiv(m, block_size_m)
114         total_blocks_n = triton.cdiv(n, block_size_n)
```

Python and ML libraries (PyTorch, tensorflow, JAX etc.)

Train a Model with Python

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Wekinator is a free, open source software that allows anyone to use machine learning to build new musical instruments, gestural game controllers, computer vision or computer listening systems, and more (Fiebrink, 2019).

www.wekinator.org

Often you want some *input* (a tone, a gesture, an image, a movement, etc.) map to a specific *output* (a tone, an action, a set of synth parameters, etc.). What you actually want is a function

$$f(input) = output.$$

Even an LLM like ChatGPT is “just” a function where the input and the output is text!

Interactive Machine Learning: The Wekinator

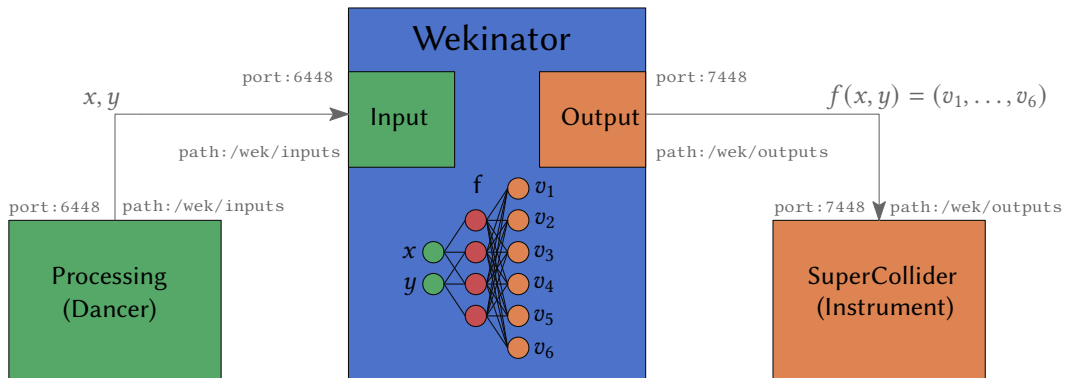
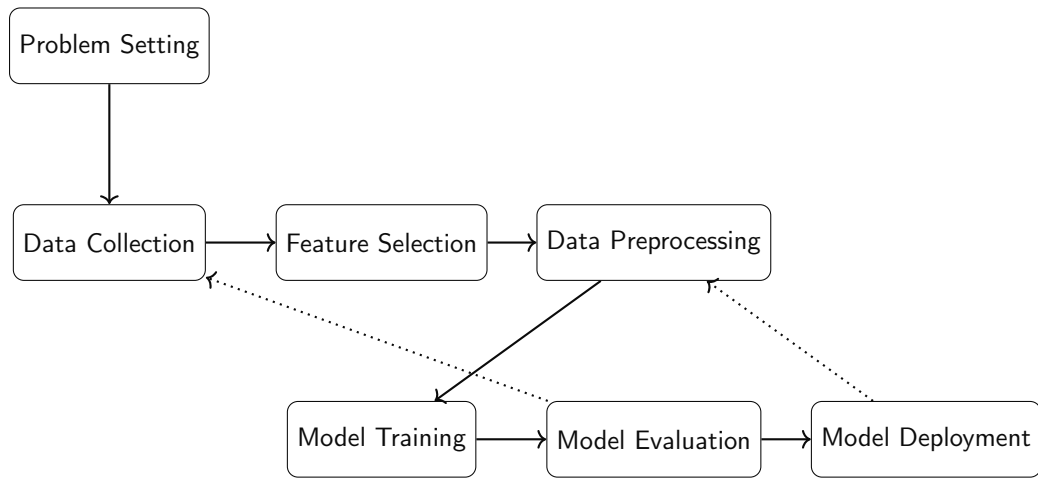


Figure: Source: Replacing Code with ML Models (blog post)

When using machine learning interactively in a creative context our goals are usually very different from using it for scientific or commercial purposes:

- We usually don't start with a "ground truth" dataset.
- Conventional evaluation metrics like model accuracy on unseen examples don't tell us enough.
- Conventional model builders focus on building the best model *of a given dataset*
 - by choosing/adjusting the learning algorithm and
 - by changing the representation of the data, i.e. feature selection.
 - But *what if we could improve the dataset itself?*
- Our models must be fast enough to respond to human input without breaking the flow of interaction.

Interactive Machine Learning: Development Cycle



Interactive Machine Learning: ML-Pipeline

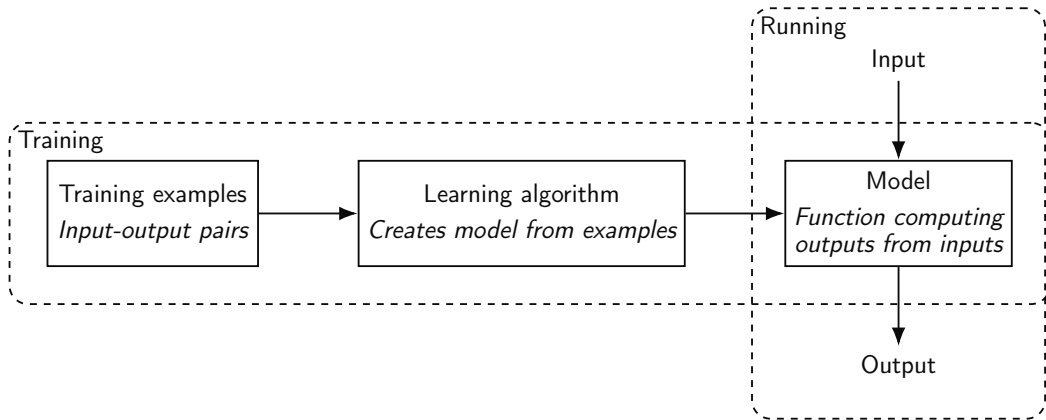


Figure: Source: (Fiebrink, 2019).

Why learning about machine learning when I don't like or don't want to code?

Paradoxically, machine learning—a technique that is built on the practice of programming—replaces, in many contexts, the need for it! Consequently, it opens up a new space of machine-human relationships that can be used and explored by everyone.

“Machine learning allows for the creation of new creative outcomes and new creative relationships between people and machines.” – Rebecca Fiebrink

ml5js is an open source library with a goal of making machine learning approachable for a broad audience of artists, creative coders, and students.

Examples

Marcelle is a software for composing interactive machine learning workflows and interfaces (Françoise, Caramiaux, & Sanchez, 2021).

<https://marcelle.dev/>

Interactive Machine Learning: The Marcelle Toolkit

Marcelle Example - Dashboard

Data Management

Training

Batch Prediction

Real-time Prediction

webcam

activate video

mobileNet

Using Mobilenet v1 with alpha = 1.

Instance label

Enter a label...

Capture instances to the training set

Hold to record instances

dataset browser

This dataset contains 65 instances.



View More

Bene



View More

Nicht Bene



View More

InteractML brings machine learning visual scripting to Unity, empowering game creators to develop novel gameplay mechanics and control schemes without writing code. You can create ML models by joining nodes together and visualize in real-time the data from a Unity scene in the graph (Bernardo, Zbyszyński, Grierson, & Fiebrink, 2020).

<https://interactml.com/>

Teachable Machine is Google's online tool which you can use to recognize sound, images, and poses.

<https://teachablemachine.com/>

Any questions?

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